

Energy performance modeling (BEM/STD)

What this tool produces · *A dynamic thermal simulation report detailing hourly heating, cooling, and electricity demands, used to validate the efficiency of the envelope and HVAC systems.*

G2 · Proof of Fit

G4 · Proof of Sustainability

Dimensions

Environmental

Spatial

Economic

Preparing the workshop · *Energy performance modeling (BEM/STD)*

G2 · Proof of Fit

Why this tool matters

Prototyping the invisible physics of shelter, allowing designers to sculpt light, wind, and heat to guarantee human comfort without sacrificing the climate.

Evidence we produce

A dynamic thermal simulation report detailing hourly heating, cooling, and electricity demands, used to validate the efficiency of the envelope and HVAC systems.

Duration & Material

Duration · 3 to 6 weeks per district/building

Material · Building Performance Simulation (BPS) software, architectural CAD files, local weather files

Reference case

The CityBES web-based platform uses EnergyPlus simulation engines to process CityGML datasets, providing city-scale building energy efficiency modeling. Municipalities use it to visualize current consumption, test large-scale retrofit scenarios, and accurately plan the deployment of district heating or renewable energy systems across thousands of varied building archetypes.

Suggested agenda · Brief (10 min) > Method walk-through (15 min) > Animation canvas (60-90 min) > Harvest (20 min)

Energy model · envelope · systems · usage · climate

Duration
3 to 6 weeks per district/building

Envelope

walls · roof · windows · airtightness

Systems

heating · cooling · ventilation · DHW

Usage

schedules · density · plug loads

Climate

site weather · UHI · microclimate

Harvesting our *learnings*

After running the canvas, capture three layers of insight.

What inspires me

Stand-out signals, surprising stories, ideas to remember

How to apply it here

Concrete adaptations to our context, our team, our constraints

What we need to be autonomous

Skills, allies, resources, decisions still to secure

Each participant fills a copy. Then debrief together to surface patterns and next steps.